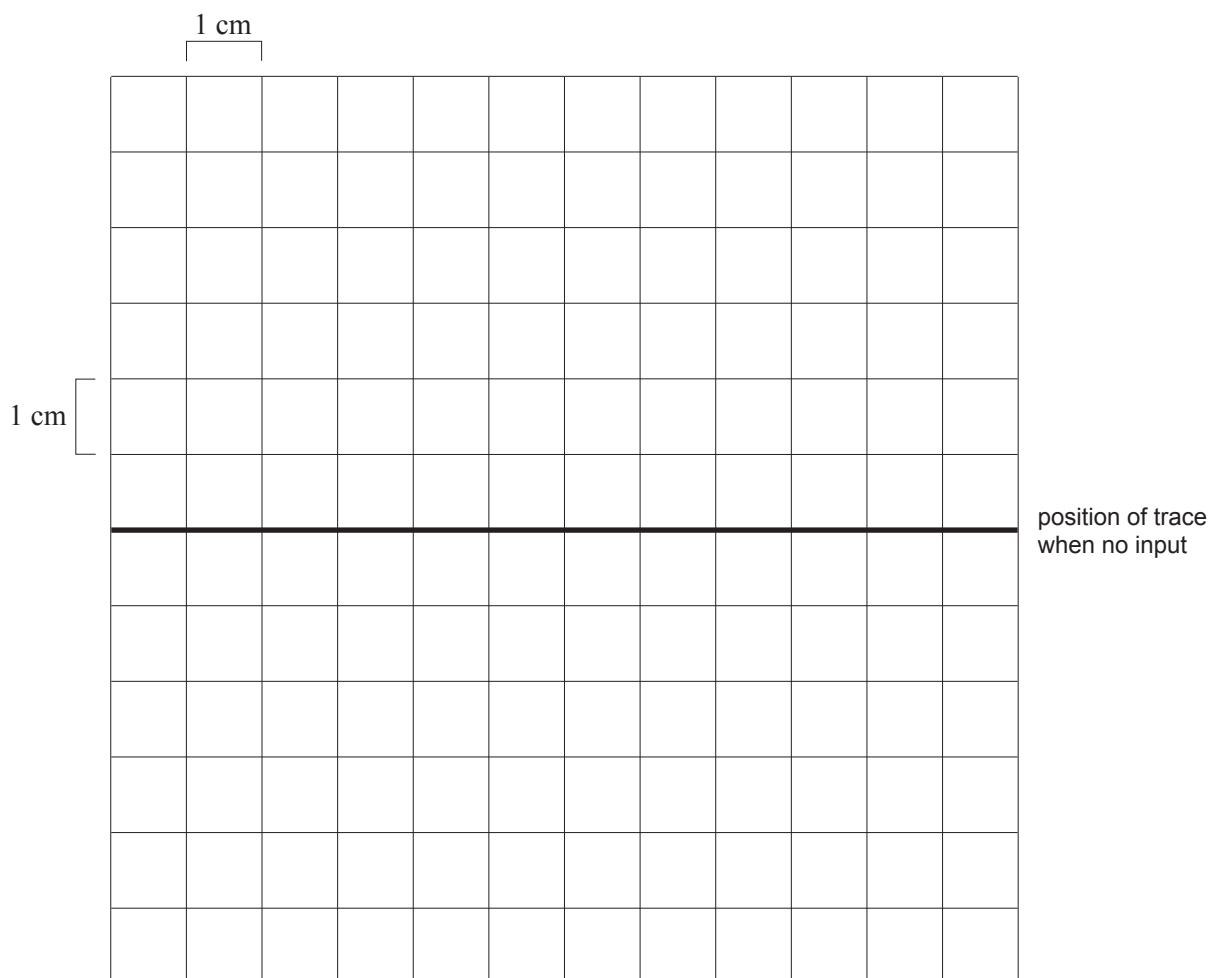


An oscilloscope is set with the vertical scale on 2 V/cm, and the horizontal scale on 5 ms/cm. With no input signal, the trace is in the position shown in Figure 6.



**Figure 6**

A  $4.0 \text{ V}_{\text{RMS}}$  sine-wave signal of frequency 50 Hz is connected to the oscilloscope.

**Question 12**

On the diagram of the oscilloscope screen shown in Figure 6, sketch the trace that would be observed.

2 marks

# **PHYSICS**

## **Written examination 1**

### **DATA SHEET**

#### **Directions to students**

Detach this data sheet before commencing the examination.

This data sheet is provided for your reference.

|    |                                                              |                                                                                                               |
|----|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 1  | velocity; acceleration                                       | $v = \frac{\Delta x}{\Delta t}; \quad a = \frac{\Delta v}{\Delta t}$                                          |
| 2  | equations for constant acceleration                          | $v = u + at$ $x = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2ax$ $x = \frac{1}{2}(v + u)t$                           |
| 3  | Newton's second law                                          | $F = ma$                                                                                                      |
| 4  | circular motion                                              | $a = \frac{v^2}{r} = \frac{4\pi^2 r}{T^2}$                                                                    |
| 5  | Hooke's law                                                  | $F = -kx$                                                                                                     |
| 6  | elastic potential energy                                     | $\frac{1}{2}kx^2$                                                                                             |
| 7  | gravitational potential energy near the surface of the Earth | $mgh$                                                                                                         |
| 8  | kinetic energy                                               | $\frac{1}{2}mv^2$                                                                                             |
| 9  | Newton's law of universal gravitation                        | $F = G \frac{M_1 M_2}{r^2}$                                                                                   |
| 10 | gravitational field                                          | $g = G \frac{M}{r^2}$                                                                                         |
| 11 | stress                                                       | $\sigma = \frac{F}{A}$                                                                                        |
| 12 | strain                                                       | $\varepsilon = \frac{\Delta L}{L}$                                                                            |
| 13 | Young's modulus                                              | $E = \frac{\text{stress}}{\text{strain}}$                                                                     |
| 14 | transformer action                                           | $\frac{V_1}{V_2} = \frac{N_1}{N_2}$                                                                           |
| 15 | AC voltage and current                                       | $V_{\text{RMS}} = \frac{1}{\sqrt{2}} V_{\text{p-p}} \quad I_{\text{RMS}} = \frac{1}{\sqrt{2}} I_{\text{p-p}}$ |
| 16 | voltage; power                                               | $V = RI \quad P = VI$                                                                                         |

|    |                                  |                                                          |
|----|----------------------------------|----------------------------------------------------------|
| 17 | resistors in series              | $R_T = R_1 + R_2$                                        |
| 18 | resistors in parallel            | $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$          |
| 19 | capacitors                       | time constant : $\tau = RC$                              |
| 20 | Lorentz factor                   | $\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$                  |
| 21 | time dilation                    | $t = t_0 \gamma$                                         |
| 22 | length contraction               | $L = L_0 / \gamma$                                       |
| 23 | relativistic mass                | $m = m_0 \gamma$                                         |
| 24 | universal gravitational constant | $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ |
| 25 | mass of Earth                    | $M_E = 5.98 \times 10^{24} \text{ kg}$                   |
| 26 | radius of Earth                  | $R_E = 6.37 \times 10^6 \text{ m}$                       |
| 27 | mass of the electron             | $m_e = 9.1 \times 10^{-31} \text{ kg}$                   |
| 28 | charge on the electron           | $q = -1.6 \times 10^{-19} \text{ C}$                     |
| 29 | speed of light                   | $c = 3.0 \times 10^8 \text{ m s}^{-1}$                   |

**Prefixes/Units**p = pico =  $10^{-12}$ n = nano =  $10^{-9}$  $\mu$  = micro =  $10^{-6}$ m = milli =  $10^{-3}$ k = kilo =  $10^3$ M = mega =  $10^6$ G = giga =  $10^9$ t = tonne =  $10^3 \text{ kg}$ **END OF DATA SHEET**